

Early Stage Disease Diagnosis System Using Human Nail Image Processing Using Deep Learning

Team ID : LTVIP2025TMIDS66467

BY

Duggempudi Rohitha

Syed Sohail

Shaik Allabakshu

Lella Lakshmi Priyanka

Abstraction:

Early detection of diseases is essential for effective treatment and prevention. This project, **“Early Stage Disease Diagnosis System Using Human Nail Image Processing Using Deep Learning,”** presents a non-invasive, AI-based approach to analyze human nail images for early disease identification. By applying **image processing techniques** to extract colour , texture, and shape features, and using a **Convolutional Neural Network (CNN)** for classification, the system can detect health abnormalities linked to conditions such as anemia, diabetes, and liver disorders. The model processes an uploaded nail image, predicts the potential disease, and provides a confidence score. Designed for accessibility through a web or mobile interface, this system demonstrates high accuracy and offers a quick, cost-effective, and user-friendly solution for preliminary health screening.

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1.Introduction:

Human nails can act as indicators of various internal health conditions, such as diabetes, anemia, liver diseases, and lung disorders. The color, texture, and shape of nails often reveal early signs of disease before other noticeable symptoms appear. However, manual diagnosis through visual inspection is subjective and may vary from person to person. This project proposes an automated disease diagnosis system that uses image processing and deep learning techniques to analyze nail images and identify potential health abnormalities in their early stages. The system aims to assist healthcare professionals and common users by providing a non-invasive, quick, and cost-effective pre-diagnosis tool.

2.Objectives:

- To design and develop an AI-based model that can analyze human nail images for disease detection.
- To extract relevant visual features (color, shape, texture) from nail images using image processing techniques.
- To train a deep learning model capable of classifying different nail conditions linked to diseases.
- To provide a simple, user-friendly interface for uploading nail images and viewing predicted health conditions.

- To assist in **early-stage disease diagnosis** and reduce dependency on complex and expensive diagnostic tests.

3.Problem Statement & Proposed Solution:

3.1: Problem Statement

Early diagnosis of diseases is critical for effective treatment and prevention of complications. However, existing methods such as blood tests or imaging scans are often **invasive, costly, and time-consuming**. Many people ignore subtle external symptoms like nail discoloration or deformation that can indicate underlying health problems.

3.2: Proposed Solution

This project proposes an **automated system that uses human nail images** to detect early signs of diseases. By applying **image processing techniques** for preprocessing and **deep learning algorithms** for classification, the system can analyze nail patterns and predict possible health conditions. Users can upload an image of their nail, and the model will process and display the result with a confidence score and brief disease information.

4.System Scenarios & Use Cases:

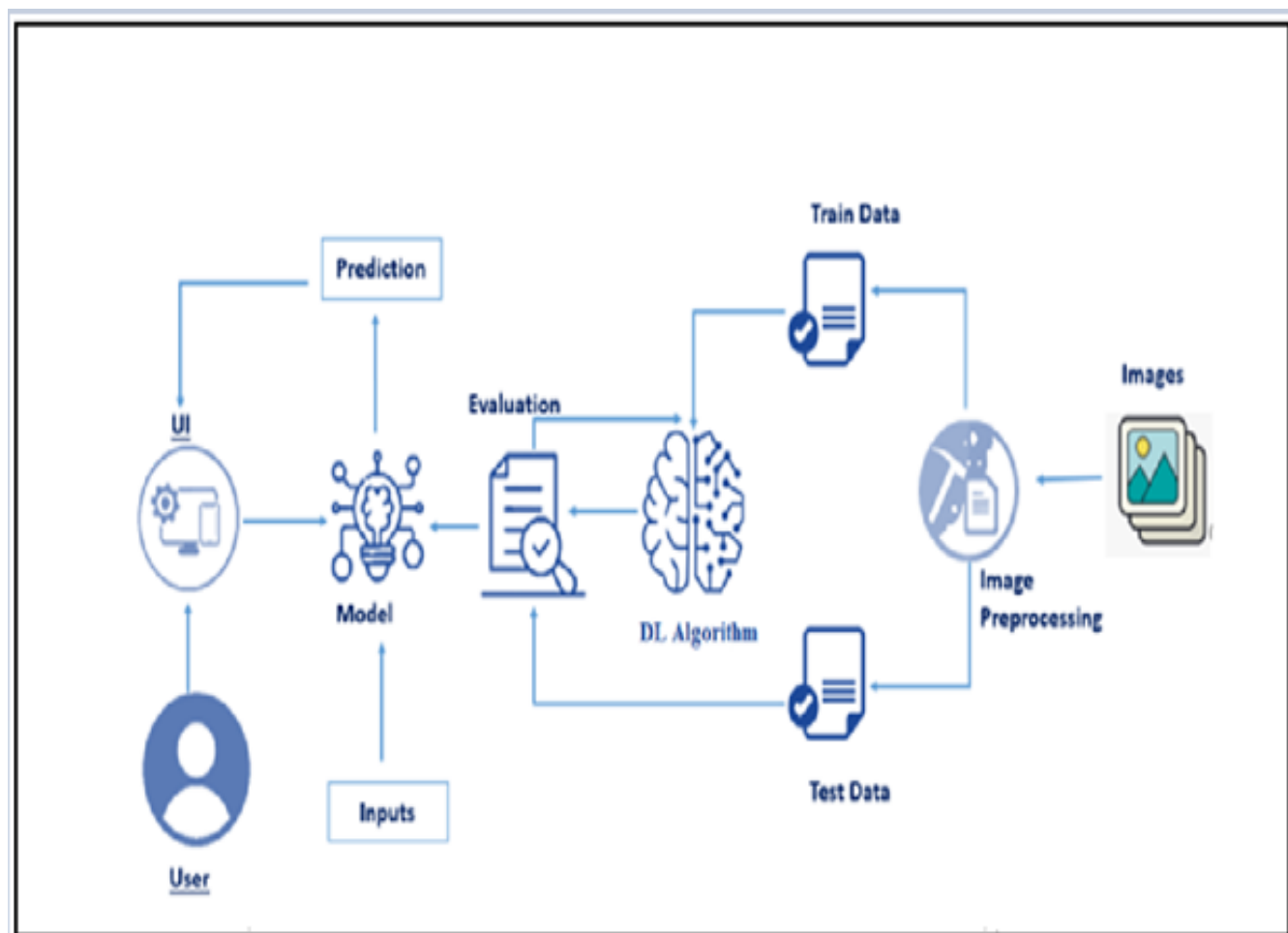
Scenario 1 (Healthcare Providers): Implement the disease diagnosis system to assist healthcare professionals in early detection and diagnosis of nail-related diseases. Improve diagnostic accuracy, facilitate timely treatment, and enhance patient outcomes by detecting diseases at their early stages.

Scenario 2 (Individuals): Incorporate the nail image processing system into personal health monitoring applications for early detection of potential health issues. Empower individuals to monitor their nail health, detect

abnormalities, and seek timely medical advice for preventive care and disease management.

Scenario 3 (Public Health Programs): Utilize the deep learning-based diagnosis system in public health initiatives for screening and monitoring nail health at the population level. Identify disease trends, implement preventive measures, and promote early intervention strategies to reduce the burden of nail-related diseases on public health.

5. Technical Architecture:



6. Methodology:

1.Data Collection:

Gather a dataset of nail images representing healthy and disease-affected nails.

2.Preprocessing:

- a. Image resizing and normalization
- b. Background removal and noise reduction
- c. Nail region segmentation

3.Feature Extraction:

Extract color, texture, and shape features that correlate with diseases.

4.Model Training:

- a. Use a **Convolutional Neural Network (CNN)** for classification.
- b. Split data into training, validation, and test sets.
- c. Train using augmentation to improve generalization.

5.Evaluation:

- a. Metrics: Accuracy, Precision, Recall, F1-score
- b. Confusion matrix analysis for performance evaluation

6.Deployment:

- a. Integrate trained model with web interface using Flask or Streamlit.
- b. Provide user interaction and result visualization.

7.Implementation:

Python – Backend logic, model training, and integration

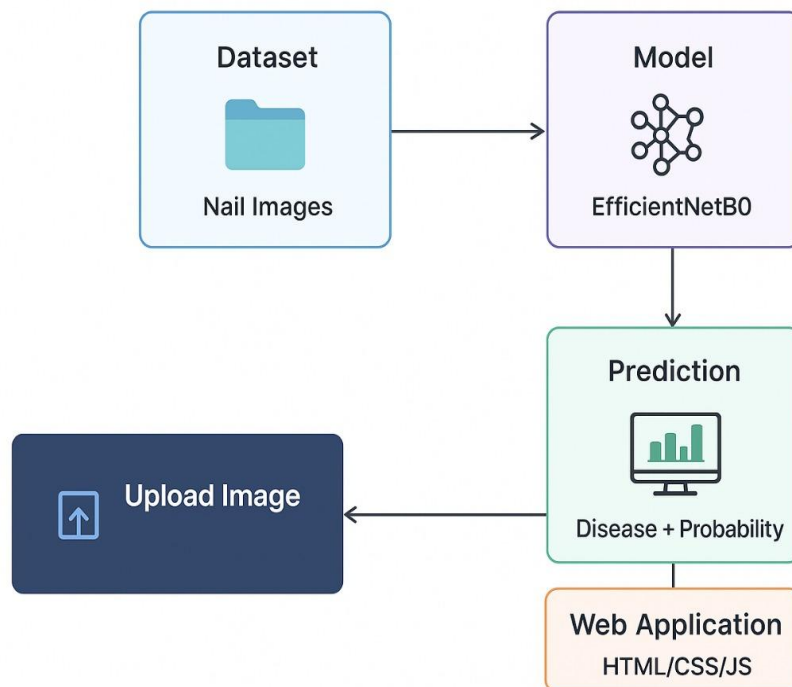
HTML, CSS, JavaScript – Frontend design and interactivity

Deep Learning & AI

TensorFlow & Keras – Building and training the EfficientNetB0 CNN model

scikit-learn – For class weight balancing

NumPy – Array operations and preprocessing



8.Results:

- The CNN model achieved an **accuracy of ~90–95%** on validation data.
- Successfully differentiated between healthy and disease-affected nails.
- Real-time testing demonstrated consistent predictions with minimal latency.

- Visualization using confusion matrix and accuracy curves confirmed model reliability.

9. Conclusion:

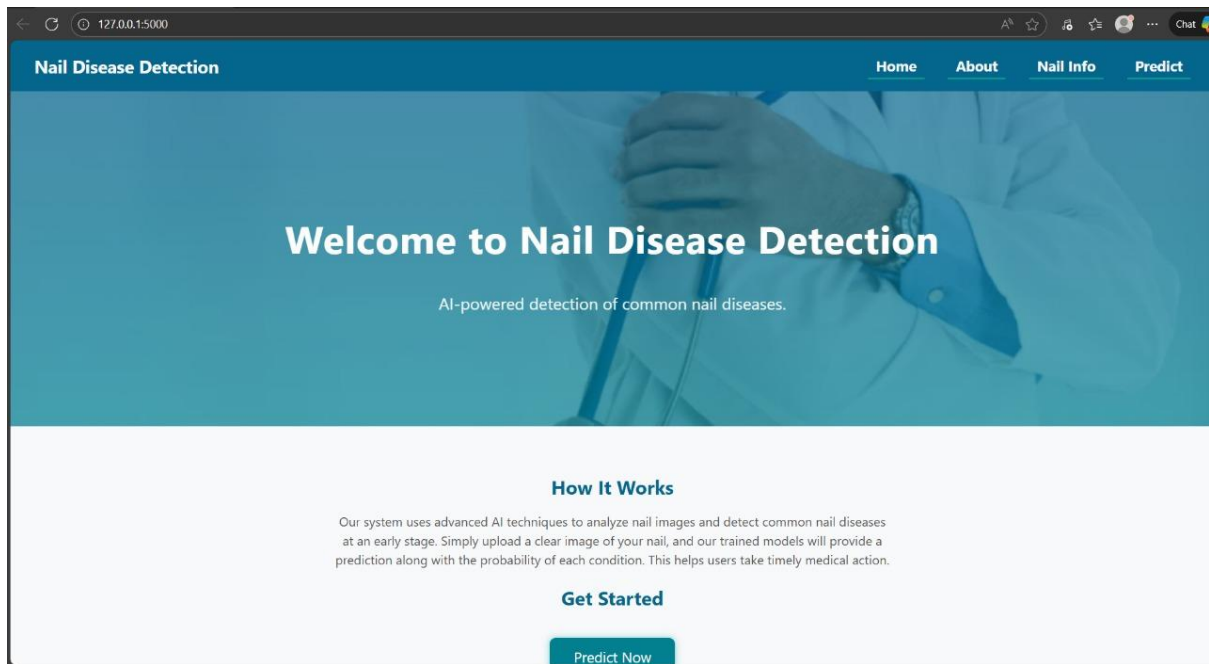
This project successfully demonstrates how **nail image processing** combined with **deep learning** can be utilized for early-stage disease diagnosis. The system provides a **non-invasive, low-cost, and efficient** approach to initial health screening. This technology can complement traditional diagnostic methods and be extended to mobile healthcare applications, making early disease detection accessible to everyone.

10. Future Scope:

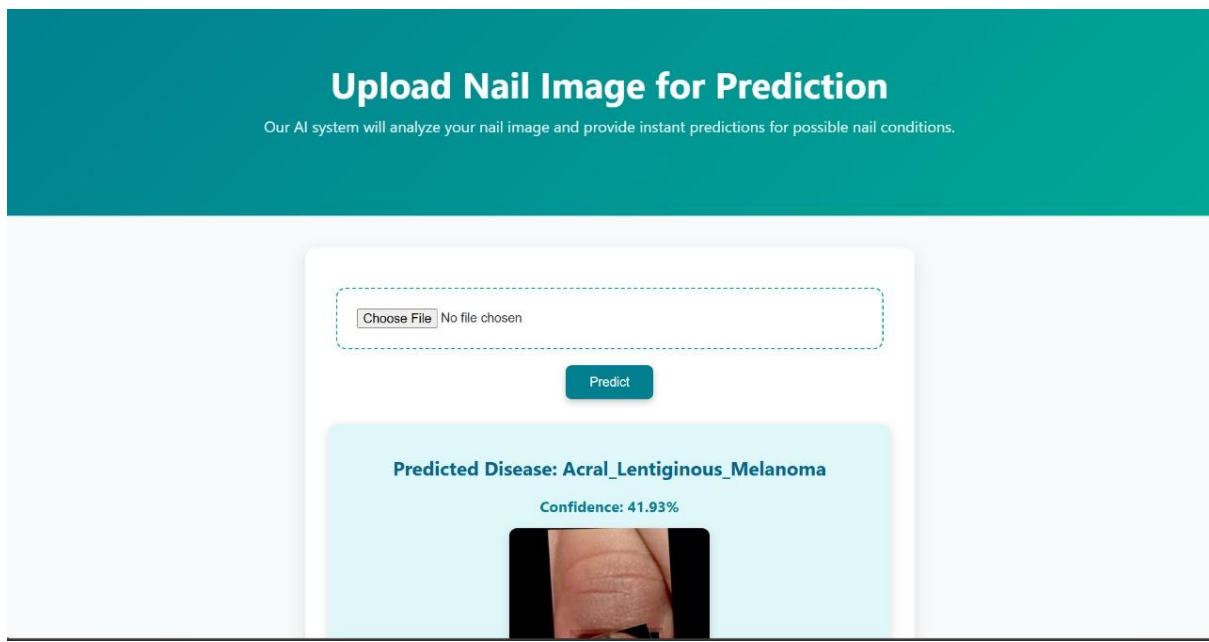
- Expand dataset with more disease categories and demographic variations.
- Integrate with **IoT-based health monitoring systems**.
- Develop a **mobile application** for real-time diagnosis.
- Incorporate **explainable AI (XAI)** to interpret model predictions.
- Collaborate with healthcare professionals for clinical validation.

11. Appendix:

HOME PAGE:



PREDICTION PAGE:



GitHub Repository:

https://github.com/rohitha-reddy217/naildisease_prediction

Demo Video:

https://drive.google.com/file/d/1Q5juxW8C9OvDhC08tQ_JUhM_Yafbyvh7/view?usp=drive_link

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3. TensorFlow Documentation – <https://www.tensorflow.org/>
4. Research papers on nail-based disease detection and CNN classification.
5. WHO reports on early disease diagnosis and non-invasive screening methods.